

Quiz 2

NAME: _____

Q.1. [5 points] Simplify the following expression (a and b can be any real numbers):

$$\frac{\sqrt{a^4b^6}}{\sqrt{a^{12}b^8}}$$

Q.2. [5 points] Fully factor the following polynomial:

$$x^2 - y^2 - 6x - 9$$

TYPO IN QUESTION (should have been $x^2 - y^2 - 6y - 9$) QUESTION CANCELLED

Q.3. [5 points] Simplify the following rational expression:

$$\frac{x^3 - 2x^2}{x^2 - 3x - 10}$$

Q.4. [5 points] Simplify the following combination of rational expressions:

$$\frac{1}{x+1} - \frac{1}{x+2} + \frac{3}{(x+1)(x+2)}$$

BONUS: Rationalize the denominator:

$$\frac{1}{\sqrt[3]{a} - \sqrt[3]{b}}$$